EMOTIVE: A Closed-Loop Affective Narrative System (CLANS) Using In-Browser Facial Blendshapes and Procedural Soundscapes

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Abstract

Interactive storytelling systems powered by Large Language Models (LLMs) predominantly operate as unidirectional generation pipelines: the reader passively consumes text, or intermittently intervenes via explicit choice trees. In this paper, we present **EMOTIVE**, a zero-install, privacy-preserving **Closed-Loop Affective Narrative System (CLANS)** that continuously and unobtrusively modulates narrative trajectory, dramatic counterpoint, and multi-layered procedural soundscapes based on real-time involuntary facial micro-expressions. Operating entirely client-side within a sandboxed WebAssembly runtime, EMOTIVE tracks 52 Facial Action Coding System (FACS) blendshapes at $\approx$ 10 Hz using MediaPipe FaceLandmarker without transmitting video frames, pixel data, or biometric templates to any server. To eliminate camera flicker, involuntary blinks, and micro-expressions from causing narrative instability, we implement an Exponential Moving Average (EMA) filter ($\alpha = 0.22$) paired with a state hysteresis barrier ($\delta = 0.09$). Rather than employing simplistic emotional mirroring (e.g., generating lighthearted content when a smile is detected), the narrative controller adopts **Psychological Counterpoint**, dynamically enriching prose with dramatic irony, bittersweet subversion, or suspenseful thresholds. In parallel, a procedural Web Audio synthesizer modulates 4-voice harmonic ambient pads and resonant biquad filters without external audio assets. We detail the mathematical signal pipeline, narrative prompt structuring, zero-biometrics privacy architecture, and an empirical telemetry protocol measuring reading dwell pace (WPM), transition stability, and post-reading Likert evaluations.

Keywords

Affective Computing, Interactive Storytelling, Large Language Models, In-Browser Computer Vision, Web Audio API, Closed-Loop Interaction, Human-Computer Interaction

1 Introduction

Generative artificial intelligence has democratized the on-demand creation of long-form literary prose. However, modern human-AI storytelling systems remain fundamentally **unidirectional**: a model generates narrative tokens, and the human reader serves merely as an optical consumer. Where interactivity is attempted—such as in classic "Choose Your Own Adventure" branching paradigms—interaction is discrete, cognitively disruptive, and relies on explicit button presses that rupture narrative absorption and literary flow [8, 11].

In live human storytelling, by contrast, a master orator never speaks in a vacuum. A storyteller constantly observes the audience's subtle bodily cues: a furrowed brow, an involuntary gasp, a leaning forward in suspense, or a fleeting smirk. These involuntary micro-expressions provide continuous affective feedback that guides the speaker's vocal cadence, tension, and thematic emphasis.

1.1 Challenges of Browser Affective Systems

Bringing closed-loop affective feedback to consumer web browsers presents three severe technical and design challenges:

(1) **Biometric Privacy & Trust**: Users rightly refuse to upload raw camera streams or facial recordings to external servers. Affective tracking must execute 100% locally on edge hardware with zero server-side image leakage.
(2) **Signal Noise & False Transitions**: Raw computer vision blendshape estimations fluctuate wildly due to head orientation, ambient lighting variations, and involuntary blinks. Feeding instantaneous raw classifications directly into generative prompts produces chaotic, unreadable narratives.
(3) **The Fallacy of Direct Mirroring**: Intuitive affective systems tend to naively mirror user emotion (e.g., if the user looks sad, tell a cheerful joke). In artistic literature, however, direct mirroring feels patronizing and breaks suspense. Compelling narrative requires **artistic counterpoint**—deepening sorrow with bittersweet solace, or introducing tension precisely when the reader feels comfortable.

1.2 Core Contributions

In this work, we present **EMOTIVE**, addressing each challenge through the following contributions:

- **Zero-Install Client-Side Vision**: An in-browser WebAssembly (WASM) vision pipeline utilizing GPU-accelerated MediaPipe FaceLandmarker to infer 52 FACS blendshapes and categorize six primary affective states (`HAPPY`, `SAD`, `SURPRISED`, `FEAR`, `ANGRY`, `NEUTRAL`) at $\approx$ 10 Hz with zero server transmission.
- **Dual-Stage Noise Rejection**: A mathematical formulation combining Exponential Moving Average smoothing ($\alpha = 0.22$) with a dynamic hysteresis boundary ($\delta = 0.09$) and observation window accumulation, eliminating blink artifacts while preserving response latency under 400 ms.
- **Psychological Counterpoint**: A structured prompting engine that steers LLM paragraph generation (Google Gemini 2.5 Flash) using literary counterpoint principles rather than

direct mirroring, maintaining strict continuity, character agency, and multi-beat emotional trajectories.
- **Procedural Web Audio Engine**: A self-contained Web Audio synthesis system that dynamically cross-fades 4-voice harmonic ambient pads and resonant biquad filter cutoffs in real time without external audio samples or network overhead.
- **Empirical Research Telemetry**: A lightweight, anonymous telemetry instrumentation system tracking word-per-minute (WPM) reading dwell times, emotional transitions, and integrated post-reading Likert scales.

2 Related Work

2.1 Affective Computing & Blendshapes

Ekman and Friesen's Facial Action Coding System (FACS) [4] remains the foundational standard for decomposing facial motion into individual Action Units (AUs). Rosalind Picard's seminal work on Affective Computing [9] established the necessity of empathetic computational interfaces. While early systems required specialized multi-spectral cameras or intrusive physiological sensors (GSR, ECG), recent advances in deep learning landmarker architectures—specifically Google's MediaPipe [6]—allow real-time 3D facial mesh estimation and 52-blendshape decomposition directly inside web browsers via WebAssembly and WebGL/WebGPU.

2.2 Interactive Narrative & LLM Steering

Computational interactive narrative generation has evolved from rule-based planners like Façade [7] to contemporary Large Language Model steering [1, 10]. While LLMs excel at generating localized prose, steering them to produce sustained literary tension across sequential beats remains challenging. Works like TaleBrush [2] explored sketch-based visual sentiment steering. However, TaleBrush relies on manual, explicit user trajectory drawing. EMOTIVE differs fundamentally by deriving steering signals **implicitly and continuously** from autonomic reader physiology.

2.3 Procedural Audio in Affective Systems

Audio significantly modulates narrative transportation and emotional arousal [3]. Traditional interactive games utilize pre-recorded audio stems with horizontal re-sequencing or vertical layering. In browser environments, pre-recorded audio assets incur substantial latency, bandwidth, and licensing constraints. Procedural synthesis using the native Web Audio API [12] enables continuous parameter modulation directly synchronized with reader affect.

3 System Architecture & Signal Pipeline

The EMOTIVE architecture operates across three distinct tiers: (1) client-side in-browser WASM perception, (2) the psychological counterpoint controller, and (3) an LLM generation loop.

3.1 In-Browser Vision & FACS Blendshapes

EMOTIVE accesses the reader's local webcam via standard `getUserMedia` constrained to 480×360 at 30 FPS. Frames are processed by `@mediapipe/tasks-vision` loaded in a WebAssembly worker with GPU acceleration. The landmarker yields 52 normalized blendshape coefficients $b_k \in [0, 1]$.

We map raw blendshapes to six primary emotional categories via empirical Action Unit combinations:

$$S_{\text{happy}} = \min(1.0,\ 1.45 \cdot \bar{b}_{\text{smile}} + 0.55 \cdot \bar{b}_{\text{cheekSquint}}) \tag{1}$$

$$S_{\text{sad}} = \min(1.0,\ 1.30 \cdot \bar{b}_{\text{frown}} + 0.85 \cdot b_{\text{browInnerUp}} + 0.35 \cdot \bar{b}_{\text{mouthPress}}) \tag{2}$$

$$S_{\text{surp}} = \min(1.0,\ 0.94 \cdot \bar{b}_{\text{eyeWide}} + 0.77 \cdot \bar{b}_{\text{browOuterUp}} + 0.94 \cdot b_{\text{jawOpen}}) \tag{3}$$

$$S_{\text{fear}} = \min(1.0,\ 0.85 \cdot \bar{b}_{\text{eyeWide}} + 0.90 \cdot b_{\text{browInnerUp}} + 0.95 \cdot \bar{b}_{\text{mouthStretch}}) \tag{4}$$

$$S_{\text{angry}} = \min(1.0,\ 1.35 \cdot \bar{b}_{\text{browDown}} + 0.70 \cdot \bar{b}_{\text{noseSneer}} + 0.65 \cdot \bar{b}_{\text{mouthPress}}) \tag{5}$$

where bilateral blendshapes are averaged: $\bar{b}_{\text{smile}} = \frac{1}{2}(b_{\text{smileL}} + b_{\text{smileR}})$.

Neutral expression probability is derived as the complementary floor:

$$S_{\text{neutral}} = \max\left(0.12,\ 1.0 - 1.15 \cdot \max_{c \in C_{\text{expressive}}} S_c\right) \tag{6}$$

The resulting vector is normalized over the unit simplex:

$$P_c(t) = \frac{S_c(t)}{\sum_{j \in C} S_j(t) + \epsilon} \tag{7}$$

3.2 Dual-Stage Signal Filtering: EMA & Hysteresis

Raw camera classifications exhibit high frequency jitter caused by lighting flicker, gaze saccades, and involuntary blinks (100–400 ms). To achieve smooth temporal integration, EMOTIVE applies a two-stage filter:

3.2.1 Stage 1: Exponential Moving Average (EMA). For each emotional category $c \in C$ at discrete time-step t:

$$\bar{P}_c(t) = \alpha \cdot P_c(t) + (1 - \alpha) \cdot \bar{P}_c(t - 1) \tag{8}$$

Empirical evaluation identified $\alpha = 0.22$ as the Pareto-optimal smoothing coefficient, providing robust blink rejection while restricting response lag to < 400 ms.

3.2.2 Stage 2: State Hysteresis Barrier. To prevent rapid oscillating transitions between competing dominant emotions, a transition from current state E_{curr} to candidate state E_{cand} occurs if and only if:

$$\bar{P}_{E_{\text{cand}}}(t) > \bar{P}_{E_{\text{curr}}}(t) + \delta_{\text{hysteresis}} \tag{9}$$

where $\delta_{\text{hysteresis}} = 0.09$. If the margin is not satisfied, the system retains E_{curr}.

4 Adaptive Narrative Engine & Psychological Counterpoint

4.1 The Counterpoint Taxonomy

Rather than naively reflecting the reader's state, EMOTIVE implements a literary **Psychological Counterpoint Engine**:

Table 1: Psychological Counterpoint Narrative Taxonomy

Sensed Affect	Naive Mirroring	Counterpoint (EMOTIVE)
HAPPY	Cheerful joke	Bittersweet nuance, poignant cost
SAD	Melodramatic grief	Solace, unexpected relic of hope
SURPRISED	Over-explanation	Narrative twist reframing prior clues
FEAR	Shock horror	Atmospheric suspense, moral choice
ANGRY	Escalated violence	Decisive defiance, cathartic agency
NEUTRAL	Random shock	Sensory grounding, atmospheric mystery

4.2 Sequential Generation & JSON Contract

EMOTIVE generates narratives incrementally, producing one self-contained paragraph (85–140 words) per narrative beat. To ensure programmatic reliability, the model is constrained to return a strict JSON schema containing: `paragraph`, `narrative_mood`, `emotional_intensity`, `plot_development`, and `continuity_notes`. The system instruction strictly prohibits the LLM from mentioning cameras, vision models, or reader physiology in the prose.

4.3 Vonnegut Emotional Arc Visualization

Following Kurt Vonnegut's narrative shape taxonomy [13], each paragraph's emotional valence $V \in [-1.0, +1.0]$ and intensity $I \in [0.0, 1.0]$ are plotted on an interactive HTML5 canvas, mapping the cumulative dramatic trajectory across the reading session.

5 Procedural Web Audio Engine

To reinforce narrative immersion without streaming pre-recorded audio files, EMOTIVE implements a procedural sound engine using the native browser Web Audio API.

5.1 Harmonic Frequency Mapping

Each affective state dynamically shifts the base frequencies of four oscillators:

- HAPPY: Cmaj9 (130.81, 196.00, 246.94, 329.63 Hz), cutoff 680 Hz.
- SAD: Am7 (110.00, 164.81, 196.00, 261.63 Hz), cutoff 360 Hz.
- SURPRISED: Dsus2 (146.83, 220.00, 329.63, 440.00 Hz), cutoff 780 Hz.
- FEAR: Diminished (87.31, 130.81, 185.00, 220.00 Hz), cutoff 260 Hz.
- ANGRY: Power fifths (82.41, 123.47, 164.81, 246.94 Hz), cutoff 520 Hz.
- NEUTRAL: Open fifths (110.00, 164.81, 220.00, 329.63 Hz), cutoff 440 Hz.

Transitions utilize `exponentialRampToValueAtTime` over a 2.8 s window, preventing clicks, acoustic pops, or abrupt tonal disruptions.

6 Anonymous Telemetry & Study Protocol

6.1 Zero-Biometrics Privacy Architecture

EMOTIVE enforces an uncompromising privacy guarantee:

(1) **Client Isolation**: Video capture and blendshape classification execute strictly within the client's WebAssembly sandbox.
(2) **Explicit Track Termination**: When the session concludes, `MediaStream` tracks are stopped, terminating camera hardware immediately.
(3) **Sanitized Telemetry**: Serverless endpoints receive only aggregated numeric scalars (dwell times, WPM reading velocity, aggregate emotion counts, and optional Likert evaluations). No IP addresses or biometric templates are stored.

6.2 Empirical Study Design & Hypotheses

To validate EMOTIVE, we formulate three formal hypotheses:

- H_1 **(Narrative Transportation)**: Closed-loop affective adaptation creates significantly higher Narrative Transportation Scale (NTS [5]) scores than static, linear reading.
- H_2 **(Affective Attunement)**: Psychological counterpoint yields higher perceived narrative depth and responsiveness than 1:1 emotional mirroring.
- H_3 **(Multi-Modal Synergy)**: Synchronous procedural audio modulates reading pace consistency, reducing rapid skimming without causing cognitive fatigue.

The telemetry collector (`/api/telemetry`) logs:

$$\text{Session} = \left\langle \text{ID},\ T_{\text{total}},\ \text{WPM}_{\text{avg}},\ \vec{C}_{\text{emotions}},\ [\Delta t_k, w_k, e_k, c_k]_{k=1}^{K},\ \vec{R}_{\text{likert}} \right\rangle \tag{10}$$

where $\vec{R}_{\text{likert}}$ records post-reading evaluations on a 5-point scale across Narrative Absorption, Affective Attunement, and Soundscape Synergy.

7 Conclusion & Future Work

We have presented **EMOTIVE**, an open-source, in-browser Closed-Loop Affective Narrative System that bridges computational literature, affective computing, and procedural audio. By combining local WebAssembly computer vision with Exponential Moving Average smoothing ($\alpha = 0.22$) and Psychological Counterpoint steering, EMOTIVE transforms passive reading into a dynamic, bi-directional aesthetic experience while preserving absolute reader privacy.

Future work will expand the system to multi-reader collective narratives, integrate eye-gaze pupil dilation tracking for cognitive load estimation, and explore long-form chapter-length story arcs.

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